

Designing & Prototyping Human-Centred System (HCI/HRI)

CariPen + CariBand AI-Assisted Triage System

System Overview

CariPen is an AI-assisted triage system consisting of a Smart Pen, temporary CariBand wristband, guided triage form, connected medical devices, and a nurse dashboard.

During registration, the patient receives the CariPen, CariBand, and guided triage form. The system automatically collects handwritten responses, the nurse–patient discussion with consent, physiological measurements, connected-device vital signs, and existing EHR information.

The AI organizes the collected information into a structured triage database containing demographics, chief complaint, medical history, symptoms, and validated vital signs. It then predicts an Emergency Severity Index (ESI) level. The nurse reviews the recommendation and makes the final triage decision.

AI Workflow

Registration

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CariPen, CariBand, and guided triage form assigned together

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Patient completes guided triage

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System automatically collects:

- Demographics
- Chief complaint
- Medical history
- Written responses
- Conversation summary
- CariBand measurements
- Blood pressure
- Heart rate
- Oxygen saturation
- Temperature
- Weight
- Existing EHR information

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AI organizes the collected information into a structured triage database
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AI predicts an ESI level
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Nurse reviews, confirms, or overrides

Canvas 1 — Problem Space

Users

Section	Patients	Triage Nurses
Characteristics	May be anxious, in pain, or unfamiliar with technology	Work under time pressure and manage several patients
Needs	Clear instructions, simple process, and minimal waiting	Accurate information, less manual documentation, and reliable support
Short-Term Goals	Complete triage quickly and communicate concerns	Review complete patient information and assign the correct ESI level
Long-Term Goals	Receive appropriate care sooner	Improve triage consistency and reduce documentation workload

System

Section	Details
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Short-Term Tasks	Guide the patient, record handwriting and spoken concerns, collect CariBand data, and receive connected vital signs
Long-Term Tasks	Build a structured triage database, support ESI prediction, and reduce documentation time
Advantages	Familiar patient interaction, automatic data collection, real-time monitoring, and EHR integration

Canvas 2 — Ethical Considerations

Area	Potential Problem	Proposed Solution
Physical Safety	Incorrect sensor readings could affect triage	Flag uncertain readings and require nurse verification
Transparency	Users may not understand the AI output	Display the confidence score and contributing factors
Data Security	Sensitive patient information is collected	Use encryption, secure access, and audit logs
Equality Across Users	Some patients may have language, literacy, visual, hearing, or physical barriers	Provide multilingual prompts, nurse-assisted mode, text, icons, high contrast, and colour-blind mode
Emotional Consideration	Patients may overtrust or fear the technology	Clearly explain that the system supports the nurse and does not make the final decision

Device Use	Patients may use the devices incorrectly	Use guided prompts, sensor checks, and staff assistance
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Canvas 3 — Design Guidelines

Guideline	Design Requirement
Advantage Guidelines	Reduce manual documentation and create more complete patient records
Ethical Guidelines	Protect privacy, maintain nurse oversight, support accessibility, and explain AI outputs
Environment Guidelines	Operate reliably in a noisy, busy emergency department
Interaction Guidelines	Use a guided form, automatic data collection, and minimal manual input
Behaviour Guidelines	Confirm completed steps, identify missing information, and alert abnormal readings
Form Guidelines	Use familiar, lightweight, professional, and disinfectable components

Canvas 4 — Robot Design MVP

Section	Details
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Where and When	Emergency Department and Observation Unit, 24 hours a day
System Role	AI-assisted triage and documentation helper
Personality	Calm, professional, and clear
Context-Based Behaviour	Prompts the next step, identifies missing data, and flags abnormal readings
Connections	EHR, AI triage model, CariPen, CariBand, and connected medical devices
Inputs	Writing, speech, touch, wearable measurements, EHR data, and clinical devices
Outputs	structured triage database, dashboard updates, alerts, LED feedback, and vibration
Before	Patient is registered and receives the CariPen, CariBand, and guided triage form
During	The system collects written, spoken, wearable, and connected-device data
After	The AI generates an ESI recommendation for nurse review

Canvas 5 — Environment

Section	Details
Where	ED registration, triage area, and Observation Unit
When	24/7, including high-volume periods
Primary Users	Patients and triage nurses
Secondary Users	Emergency physicians, Observation Unit staff, administrators, IT staff, and biomedical engineers
Data Collection	Demographics, chief complaint, medical history, written responses, conversation summary, vital signs, and waiting time
Simultaneous Users	One patient per CariPen and CariBand set; several sets may operate at the same time
External Sensors	Blood pressure cuff, pulse oximeter, thermometer, and weight scale
System Connections	EHR, hospital network, AI model, and connected devices

Canvas 6 — Form

Section	Details
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Appearance	Minimal, professional, familiar, and medical
Size	Standard-sized pen and lightweight temporary wristband
Movement	Handheld and wearable; no autonomous movement
Voice and Sounds	Spoken patient input and a soft confirmation tone
Mobility	Portable CariPen and wearable CariBand
Visual Cues	LED status indicators and dashboard notifications
Accessibility	Colour-blind mode, text labels, icons, high contrast, and multilingual instructions
Touch and Materials	Comfortable rubber grip, medical silicone band, and disinfectable surfaces
Sanitization	Charging and enclosed sanitizing dock between patients

Canvas 7 — Interaction

Section	Details
Input Modalities	Writing, voice, touch, physiological measurements, EHR data, and connected devices

Output Modalities	Dashboard, alerts, text, icons, LED lights, vibration, and optional sound
Situation Flow	Guided but flexible
Leadership	Shared: the system guides data collection, while the nurse controls the clinical decision
Goal	Collect complete patient information and support safe triage
System Name	CariPen and CariBand
Before	Patient registration and device assignment
During	The guided form collects written, spoken, wearable, and vital-sign information
After	The AI produces an ESI recommendation, and the nurse confirms or overrides it

System Requirements

Inputs

- Patient demographics
- Guided triage-form responses
- Handwritten patient information
- Spoken patient concerns
- AI-generated discussion summary
- Existing EHR information
- CariBand physiological measurements
- Blood pressure
- Heart rate

- Oxygen saturation
- Temperature
- Weight

Outputs

- Structured triage database
- ESI prediction
- Confidence score
- Contributing factors
- Missing or unreliable data warnings
- Critical alerts
- Audit log

Functional Requirements

The system will:

- Link the CariPen, CariBand, triage form responses, and connected devices to the correct patient.
- Digitize handwritten information.
- Record and summarize the triage discussion with patient consent.
- Organize collected information into structured fields.
- Integrate with the hospital EHR.
- Generate an ESI recommendation.
- Display the recommendation on the nurse dashboard.
- Allow the nurse to confirm, request more information, or override the recommendation.
- Record predictions and nurse decisions in an audit log.

Non-Functional Requirements

The system will be:

- Secure and encrypted
- Reliable in a busy clinical environment
- Fast enough to update the dashboard in near real time
- Accessible to users with different needs
- Compatible with a colour-blind display setting
- Easy to clean and disinfect
- Able to store information temporarily during a network failure
- Clear about missing or unreliable data
- Designed so the nurse always makes the final triage decision